

Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 3 / 12

Maximum Likelihood Estimation: A Worked Example

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1 Maximum Likelihood Estimation: Worked Examples

This chapter works through MLE explicitly in two contrasting families: the regular $\mathcal{N}(\mu, 1)$ family, where the MLE is found by setting the score to zero, and the non-regular $U[0, \theta]$ family, where the MLE is the sample maximum.

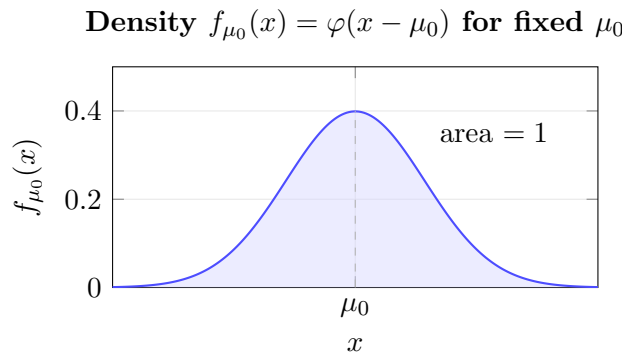
1.1 Worked example: $\mathcal{N}(\mu, 1)$ from end to end

We close the Score and Fisher Information chapter (Part 2) by working a single family — the unit-variance location-normal $\mathcal{N}(\mu, 1)$ — through every construction of the last three chapters: density, pre-likelihood, log-likelihood, the randomisation of the index, the pre-score, the score, and the integral identity $\mathbb{E}[s] = 0$. The example is small enough to compute explicitly, so the abstract pattern becomes concrete. These notes follow the Probabilist's worksheet of May 15, 2026.

Step 1. The density family. Take $\theta = \mu \in \mathbb{R}$, scalar. The density is

$$f_{\mu}(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2}\right) = \varphi(x - \mu),$$

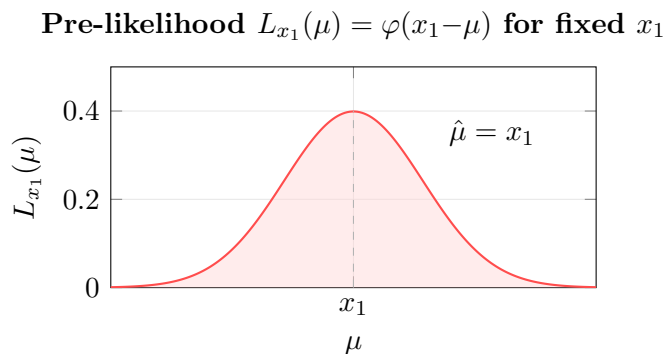
where φ denotes the standard normal density. For a fixed value μ_0 , f_{μ_0} is a bell curve in x centred at μ_0 , of unit area.



Step 2. The pre-likelihood. For a fixed observation x_1 ,

$$L_{x_1}(\mu) = \varphi(x_1 - \mu)$$

is the pre-likelihood as a function of μ . It is also a bell curve, now along the μ -axis and centred at $\mu_0 = x_1$. This is the location-normal coincidence (the location-normal symmetry remark (Part 2)): the density slice and the pre-likelihood slice look the same.

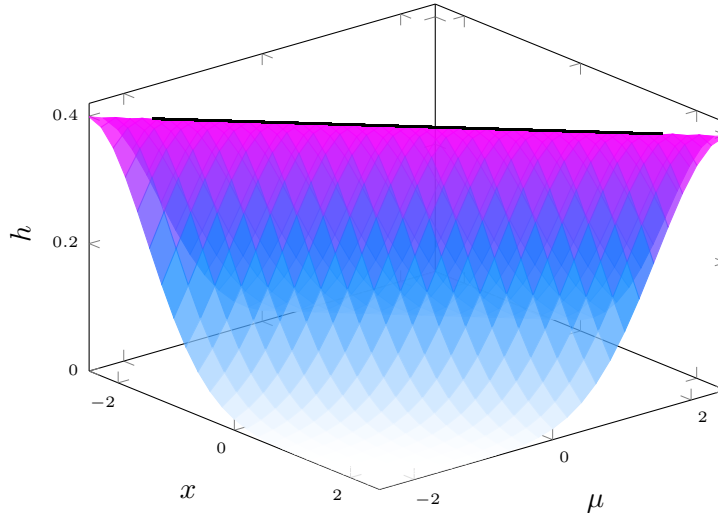


Step 3. The bivariate surface. The two slices live on one surface

$$h(x, \mu) = \varphi(x - \mu),$$

which depends on its two arguments only through the difference $x - \mu$. The ridge of the surface is the line $x = \mu$.

Bivariate surface $h(x, \mu) = \varphi(x - \mu)$; **ridge** $x = \mu$ (black)

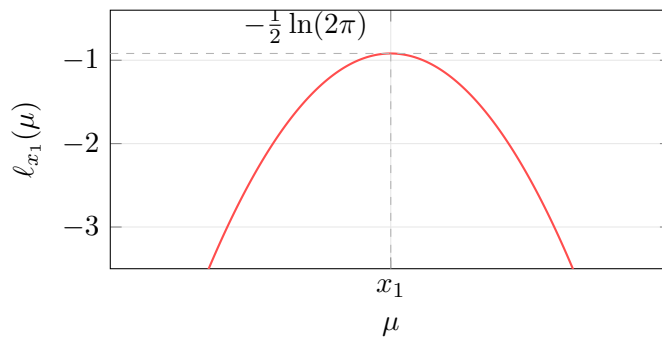


Step 4. The pre-log-likelihood. Take the log on the positivity set (all of $\mathbb{R}$ here, since $f_\mu > 0$ everywhere):

$$\ell_x(\mu) = \log h(x, \mu) = -\frac{1}{2} \log(2\pi) - \frac{1}{2}(x - \mu)^2.$$

A downward parabola in μ with vertex at $\mu = x$ and peak value $-\frac{1}{2} \log(2\pi)$ (the same value at every observation x — a peculiarity of the location-normal family).

Log-likelihood $\ell_{x_1}(\mu) = -\frac{1}{2}(x_1 - \mu)^2 - \frac{1}{2} \ln(2\pi)$, $x_1 = 0$



Step 5. Connecting Ω : random parabolas. Now treat the observation as a random variable $X(\omega) \sim \mathcal{N}(\mu_0, 1)$ for some true parameter $\mu_0 \in \mathbb{R}$. Each ω produces one realised observation $x = X(\omega)$ and hence one realised pre-log-likelihood — a single downward parabola with vertex at that observed x . Sampling many ω 's materialises a *cloud of random parabolas*, each centred at its own $X(\omega)$.

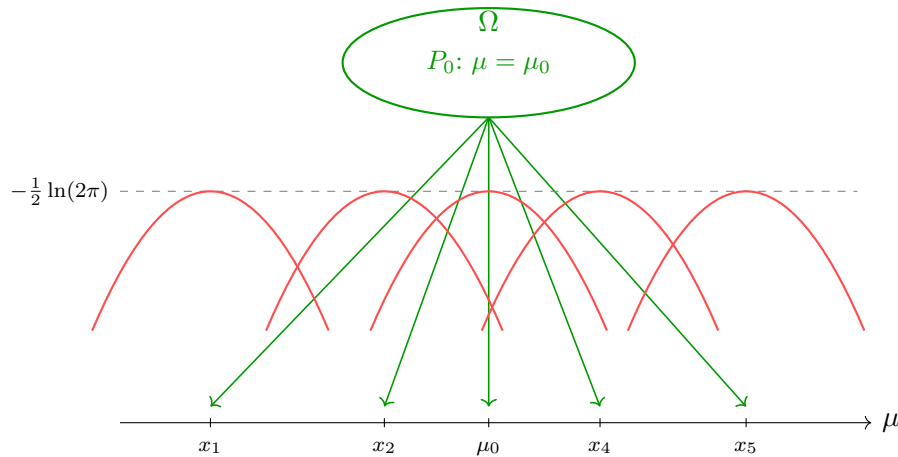
The parabolas are denser where the observations are denser. Since $X \sim \mathcal{N}(\mu_0, 1)$, the observations cluster around μ_0 and thin out in the tails, so the vertices of the random parabolas cluster around μ_0 and thin out in the tails as well. This is the picture on page 2 of the worksheet: an Ω -cloud on top,

arrows pointing down to a handful of realised x_i 's along the μ -axis, and a fan of downward parabolas with vertices at those x_i 's. The randomness of the log-likelihood is exactly this cloud of parabolas indexed by ω .

Remark 1 (The MLE for a single observation). For one observation x , $\ell_x(\mu) = -\frac{1}{2}(x - \mu)^2 + C$ is maximised at $\hat{\mu} = x$. So

$$\hat{\mu}_{\text{MLE}}(X(\omega)) = X(\omega).$$

The MLE is the observation itself. This is the simplest possible estimator and a sanity check: in this family, the data *is* the maximum-likelihood guess for the location.

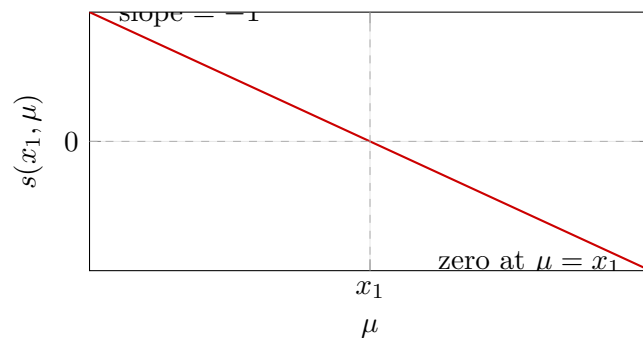


Step 6. The pre-score. Differentiate ℓ_x in μ :

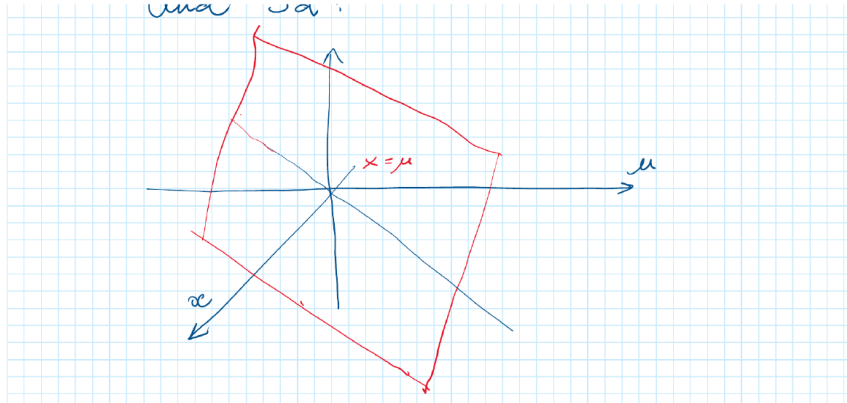
$$s(x, \mu) = \frac{\partial}{\partial \mu} \ell_x(\mu) = \frac{\partial}{\partial \mu} \left[-\frac{1}{2}(x - \mu)^2 \right] = (x - \mu).$$

The pre-score is the affine function $(x, \mu) \mapsto x - \mu$. At any fixed x_1 , it is a downward-sloping line in μ , passing through zero at $\mu = x_1$. At any fixed μ_0 , it is an upward-sloping line in x , passing through zero at $x = \mu_0$.

Pre-score $s(x_1, \mu) = x_1 - \mu$ vs. μ , for fixed $x_1 = 0$



The same affine form $s(x, \mu) = x - \mu$ as a function of two arguments describes a tilted plane in 3D: a saddle-like ramp that crosses zero along the ridge $x = \mu$, positive on one side, negative on the other. The 3D picture on page 4 of the worksheet shows this plane intersecting the zero plane along the MLE ridge $x = \mu$.



Step 7. The integral identity $\mathbb{E}_{\mu_0}[s] = 0$, **geometrically.** Now fix the true parameter μ_0 and look along the x -axis at fixed $\mu = \mu_0$. On this slice:

- the density is $f_{\mu_0}(x) = \varphi(x - \mu_0)$, a bell centred at μ_0 ;
- the score is $s(x, \mu_0) = x - \mu_0$, a straight line of slope 1 crossing zero at $x = \mu_0$;
- their product is $s(x, \mu_0) \cdot f_{\mu_0}(x) = (x - \mu_0) \varphi(x - \mu_0)$, an odd function around μ_0 .

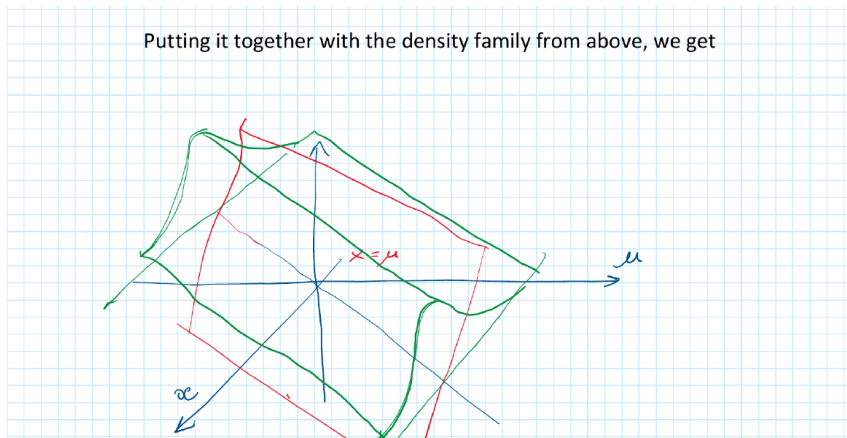
The expected score is the integral of this product:

$$\mathbb{E}_{\mu_0}[s(\mu_0)] = \int_{\mathbb{R}} s(x, \mu_0) f_{\mu_0}(x) dx = \int_{\mathbb{R}} (x - \mu_0) \varphi(x - \mu_0) dx = 0,$$

by symmetry: positive and negative areas cancel exactly.

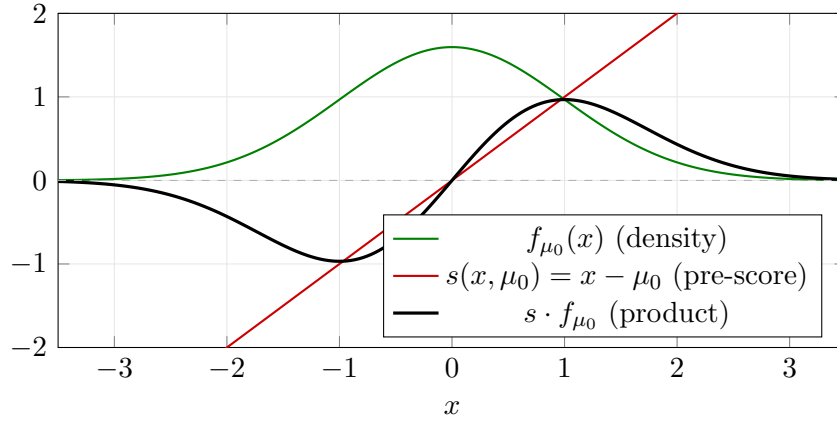
This is the diagram on page 5 of the worksheet: **green** density, **red** score (the straight line crossing zero at μ_0), **black** their product (an odd S-curve through μ_0). The black curve integrates to zero in Lebesgue measure.

The bivariate view below shows both surfaces simultaneously: the green density ridge $h(x, \mu)$ and the red pre-score plane $s(x, \mu)$ share the ridge $x = \mu$, where the score is zero and the density is at its maximum. Back-projecting to the fixed slice $\mu = \mu_0$ (the vertical cut) gives the two curves whose product is the black odd curve in the next plot.



Back to 2D. Fixing $\mu = \mu_0$ and projecting onto the x -axis yields the three curves below.

Pre-score $s(x, \mu_0)$, density $f_{\mu_0}(x)$, and their product, at $\mu_0 = 0$



Note on the rescaling: the density f_{μ_0} and the product $s \cdot f_{\mu_0}$ have been multiplied by 4 in this plot so that all three curves sit on a comparable vertical scale. The shape of each curve, and the symmetry that makes the product integrate to zero, are not affected.

By symmetry of the $\mathcal{N}(\mu_0, 1)$ density around μ_0 , this cancellation holds for *every* $\mu_0 \in \mathbb{R}$, not just $\mu_0 = 0$. The score identity $\mathbb{E}_\mu[s(\mu)] = 0$ is not a deep theorem for this family — it is an immediate visual consequence of the symmetry of the bell curve.

Step 8. The same identity as integration that cancels the denominator. There is a parallel route to the same identity that is worth flagging, since it generalises beyond the symmetric case. The pre-score is

$$s(x, \mu) = \frac{\partial_\mu h(x, \mu)}{h(x, \mu)}.$$

Taking the expectation against the density $f_\mu = h(\cdot, \mu)$ multiplies the pre-score by h in the integrand, and the denominator cancels:

$$\mathbb{E}_\mu[s(\mu)] = \int s(x, \mu) h(x, \mu) dx = \int \frac{\partial_\mu h(x, \mu)}{h(x, \mu)} \cdot h(x, \mu) dx = \int \partial_\mu h(x, \mu) dx = 0.$$

The final step is Property 3 of the Density and Likelihood chapter (Part 2). So the cancellation is not specific to the symmetry of the normal: it follows from the fact that h integrates to one in x for every μ . The normal case just gives a particularly transparent picture in which both arguments — symmetry and Property 3 — yield the same answer.

What this example gave us. A single family, computed step by step, instantiates everything from the Density/Likelihood and Score chapters (Part 2): the bivariate surface, two kinds of slices, log transform, random indexing via ω producing a cloud of parabolas, the affine pre-score $x - \mu$, the saddle-shaped score surface, and the integral $\mathbb{E}[s] = 0$ realised as the area under an odd black curve. The next subsection extends the same example to $n > 1$ observations, where the sample mean $\bar{x}$ enters as a sufficient statistic and the score acquires its characteristic $\bar{x} - \mu$ form.

1.2 Worked example continued: n observations

We continue the $\mathcal{N}(\mu, 1)$ example to the case of a sample $\mathbf{x} = (x_1, \dots, x_n)$ of size $n > 1$, i.i.d. from $\mathcal{N}(\mu, 1)$. The aim is twofold: to see how the bivariate surface picture extends to dimension $n + 1$; and to verify the identity $\mathbb{E}_\mu[s] = 0$ by direct calculation, in three rounds of increasing complexity, following the Probabilist's worksheet of May 16, 2026.

Step 9. The joint density and joint log-likelihood. By independence,

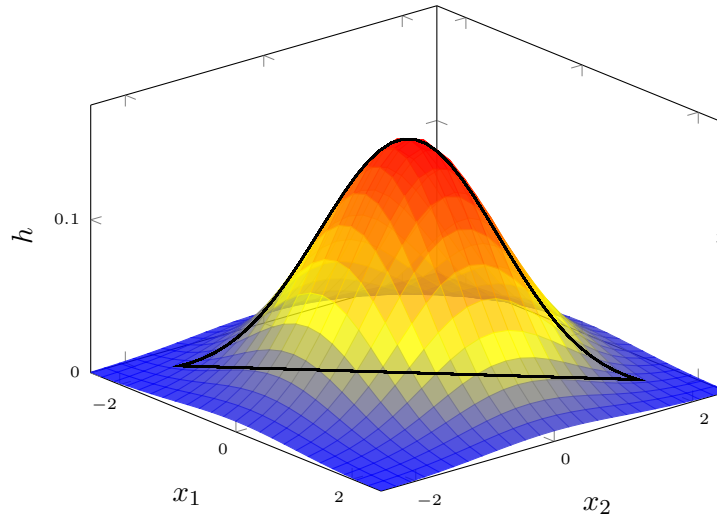
$$h(x_1, \dots, x_n; \mu) = \prod_{i=1}^n \varphi(x_i - \mu) = \frac{1}{(2\pi)^{n/2}} \exp\left(-\frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

Taking the log,

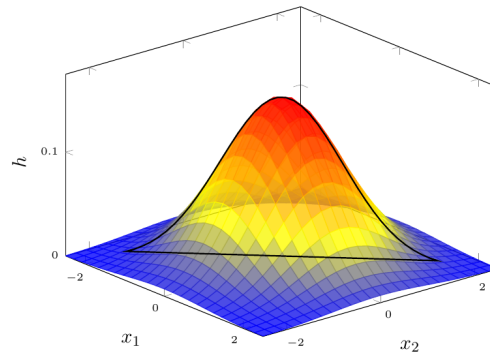
$$\ell(x_1, \dots, x_n; \mu) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2.$$

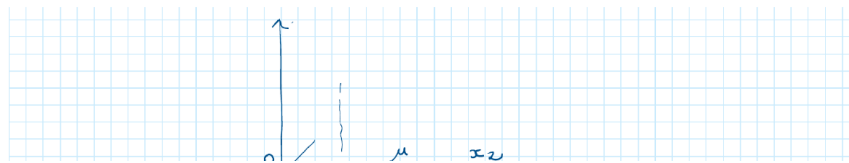
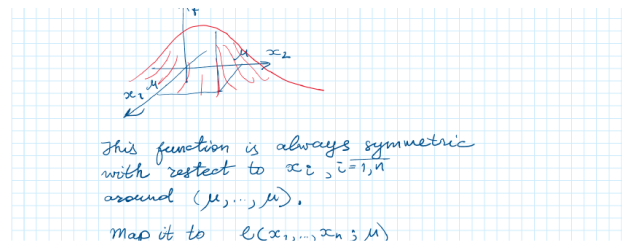
The surface in $n + 1$ dimensions. For $n = 2$, $\ell(x_1, x_2; \mu)$ is a function of three variables, hence a hypersurface in $\mathbb{R}^4$. Sectioning along the μ -axis at fixed (x_1, x_2) gives a downward parabola in μ , just as in the $n = 1$ case, but now with vertex at the sample mean $(x_1 + x_2)/2$. Sectioning along the (x_1, x_2) -plane at fixed μ gives a downward paraboloid centred at (μ, μ) ; the joint log-likelihood is rotationally symmetric in the data variables around the diagonal point $(\mu, \dots, \mu)$.

Joint density $h(x_1, x_2; \mu) = \varphi(x_1 - \mu)\varphi(x_2 - \mu)$, **ridge** $x_1 = x_2 = \mu$ (**black**), $\mu = 0$



Joint density $h(x_1, x_2; \mu) = \varphi(x_1 - \mu)\varphi(x_2 - \mu)$, **ridge** $x_1 = x_2 = \mu$ (**black**), $\mu = 0$



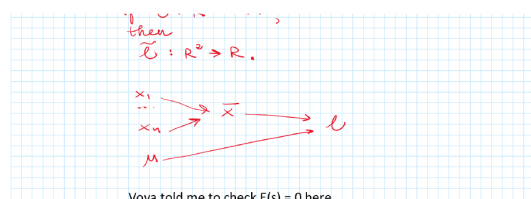
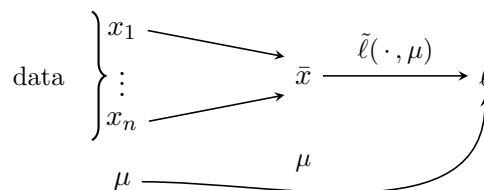


Step 10. Differentiating in μ : the sample mean appears. Differentiate the joint log-likelihood:

$$s(\mathbf{x}, \mu) = \frac{\partial}{\partial \mu} \ell(\mathbf{x}, \mu) = -\frac{1}{2} \sum_{i=1}^n \frac{\partial}{\partial \mu} (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \mu) = n\bar{x} - n\mu = n(\bar{x} - \mu),$$

where $\bar{x} = \frac{1}{n} \sum_i x_i$ is the sample mean. Two observations are worth flagging.

- **The pre-score is additive over the sample.** The joint pre-score is the sum of the individual pre-scores $s(x_i, \mu) = x_i - \mu$. This is the operator-algebra of D applied to the product $h = \prod \varphi(x_i - \mu)$: multiplication of densities becomes addition of pre-scores (Property 1 of the semi-elasticity section (Part 2)).
- **The pre-score factors through $\bar{x}$.** The joint pre-score $n(\bar{x} - \mu)$ depends on the data only through $\bar{x}$. Equivalently, $\ell(\mathbf{x}, \mu)$ can be written as $\tilde{\ell}(\bar{x}(\mathbf{x}), \mu)$ for some function $\tilde{\ell} : \mathbb{R}^2 \rightarrow \mathbb{R}$. The sample mean is a *sufficient statistic* for μ in this family. This is the entry point to sufficiency theory; we do not pursue it here.



Remark 2 (The role of μ). At this stage μ is a free variable, not the true parameter. We are differentiating ℓ with respect to μ as a function of two arguments; the result $s(\mathbf{x}, \mu) = n(\bar{x} - \mu)$ is the pre-score, a function of both. When we later substitute a *specific* value $\mu = \mu_0$ (the true parameter) and treat $\mathbf{x}$ as random under $\mathcal{N}(\mu_0, 1)$, the same expression becomes a random variable whose distribution we will compute.

Step 11. The score identity $\mathbb{E}_\mu[s] = 0$, in three rounds. We verify $\mathbb{E}_\mu[s(\mathbf{X}, \mu)] = 0$ directly by integration, in three rounds of increasing complexity.

Round 1: $n = 1$, general parametric family. Starting from the definition $s = \partial_\theta \log f_\theta = \partial_\theta f_\theta / f_\theta$ and using Path B of the semi-elasticity operator,

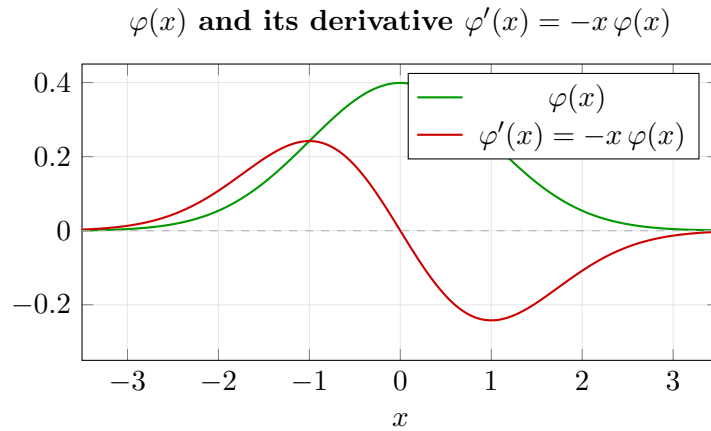
$$\mathbb{E}_\theta[s(X, \theta)] = \int_{\mathbb{R}} s(x, \theta) f_\theta(x) dx = \int \frac{\partial_\theta f_\theta(x)}{f_\theta(x)} \cdot f_\theta(x) dx = \int \partial_\theta f_\theta(x) dx = 0,$$

the last equality being Property 3 of the Density and Likelihood chapter (Part 2). The denominator f_θ in the score cancels against the density used to take expectation. This is the general route; it works for any sufficiently regular family.

Round 2: $n = 1$, normal family — explicit calculation. For $\mathcal{N}(\mu, 1)$, work the same identity by direct substitution. First a side calculation: the derivative of φ in its argument is

$$\varphi'(x) = \frac{1}{\sqrt{2\pi}} \frac{d}{dx} e^{-x^2/2} = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot (-x) = -x \varphi(x).$$

So φ' is the reflected curve $-x \varphi(x)$ — positive for $x < 0$, negative for $x > 0$.



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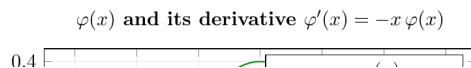
$$\mathbb{E}_\theta[s(X, \theta)] = \int_{\mathbb{R}} s(x, \theta) f_\theta(x) dx = \int \frac{\partial_\theta f_\theta(x)}{f_\theta(x)} \cdot f_\theta(x) dx = \int \partial_\theta f_\theta(x) dx = 0,$$

st equality being Property 3 of the Density and Likelihood chapter (Part 2). The denominator the score cancels against the density used to take expectation. This is the general route for any sufficiently regular family.

Round 2: $n = 1$, normal family — explicit calculation. For $\mathcal{N}(\mu, 1)$, work the same identity by direct substitution. First a side calculation: the derivative of φ in its argument is

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is the reflected curve $-x \varphi(x)$ — positive for $x < 0$, negative for $x > 0$.



Then for $\varphi_\mu(x) = \varphi(x - \mu)$, the chain rule in μ gives

$$\frac{\partial}{\partial \mu} \varphi(x - \mu) = \varphi'(x - \mu) \cdot \frac{\partial}{\partial \mu}(x - \mu) = -(x - \mu) \varphi(x - \mu) \cdot (-1) = (x - \mu) \varphi(x - \mu).$$

Taking the expectation,

$$\mathbb{E}_\mu[s(X, \mu)] = \int (x - \mu) \varphi(x - \mu) dx = \int_{-\infty}^{\infty} y \varphi(y) dy = 0,$$

by the substitution $y = x - \mu$ and the symmetry of the standard normal. The score identity is realised as the first moment of a standard normal being zero — a clean concrete answer.

Round 3: $n > 1$, normal family — factorisation. For the i.i.d. sample,

$$h(x_1, \dots, x_n; \mu) = \prod_{i=1}^n \varphi(x_i - \mu),$$

and differentiating in μ gives

$$\frac{\partial h}{\partial \mu} = \sum_{i=1}^n (x_i - \mu) \varphi(x_i - \mu) \prod_{j \neq i} \varphi(x_j - \mu).$$

(Here we are computing $\partial_\mu h$ to use Path B; equivalently one can compute $\partial_\mu \ell = n(\bar{x} - \mu)$ and multiply back by h .) Taking the expectation under $\mathcal{N}(\mu, 1)^n$:

$$\mathbb{E}_\mu[s(\mathbf{X}, \mu)] = \int \cdots \int \sum_{i=1}^n (x_i - \mu) \prod_{j=1}^n \varphi(x_j - \mu) dx_1 \cdots dx_n.$$

By independence the multidimensional integral factorises into a sum of products. For the i -th term, the integrals over x_j with $j \neq i$ each give 1 (densities integrate to one), and the integral over x_i is $\int (x_i - \mu) \varphi(x_i - \mu) dx_i = 0$ by Round 2. Every term in the sum vanishes individually:

$$\mathbb{E}_\mu[s] = \underbrace{0 + 0 + \cdots + 0}_{n \text{ terms}} = 0.$$

What this verification shows. The score identity holds at all three levels — general parametric, specific normal $n = 1$, specific normal $n > 1$ — by different but parallel arguments. The general argument is the denominator cancellation (Round 1). The specific normal $n = 1$ argument is the symmetry of the bell (Round 2). The $n > 1$ argument is the factorisation due to i.i.d. structure, reducing to n copies of Round 2 (Round 3).

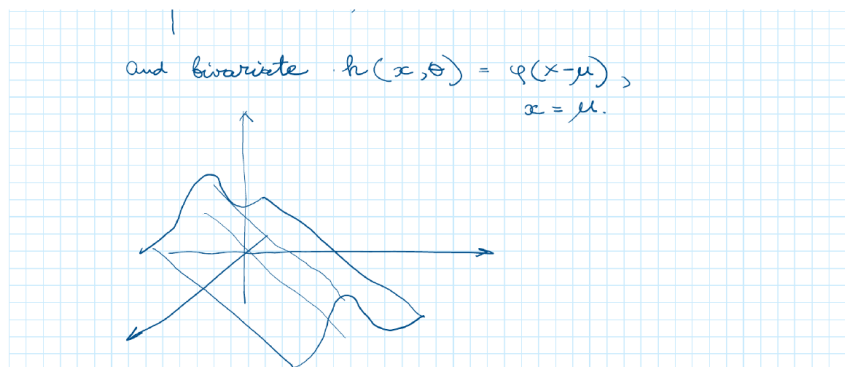
What we have now built. At the end of the Score and Fisher Information chapter (Part 2), the $\mathcal{N}(\mu, 1)$ example has been carried through every construction: density, pre-likelihood, log-likelihood, the random indexing via ω , the cloud of parabolas in finite samples, the pre-score factoring through $\bar{x}$, sufficiency, and the score identity verified three ways. The same machinery applied to test $H_0 : \mu = \mu_0$ in three different ways (Wald, LR, score) in the next chapter, and explains why they asymptotically agree.

1.3 MLE for $\mathcal{N}(\mu, 1)$: finding the argmax

We now apply the machinery of the previous two subsections to find the MLE explicitly. The question is: given data, which value of μ maximises the likelihood? We answer it first for $n = 1$, where the geometry is transparent, and then for general n , where the sample mean emerges as a sufficient reduction.

Case $n = 1$. Fix $x = x_1$ and maximise $h(x_1, \mu) = \varphi(x_1 - \mu)$ over μ . The bivariate surface $h(x, \mu)$ was drawn in Step 3; the MLE is the point on that surface directly above the diagonal ridge $x = \mu$. Cutting the surface with the vertical plane $x = x_1$ (the *red slice* in the figure below) gives the pre-likelihood $L_{x_1}(\mu) = \varphi(x_1 - \mu)$ as a function of μ alone.

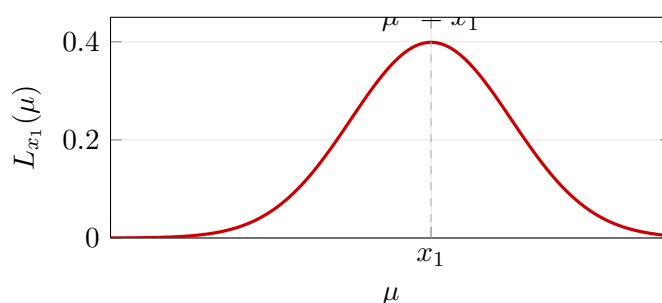
was drawn in Step 3, the MLE is the point on that surface directly above the diagonal ridge $x = \mu$. Cutting the surface with the vertical plane $x = x_1$ (the *red slice* in the figure below) gives the pre-likelihood $L_{x_1}(\mu) = \varphi(x_1 - \mu)$ as a function of μ alone.



The red slice is itself a bell curve in μ , centred at $\mu = x_1$.

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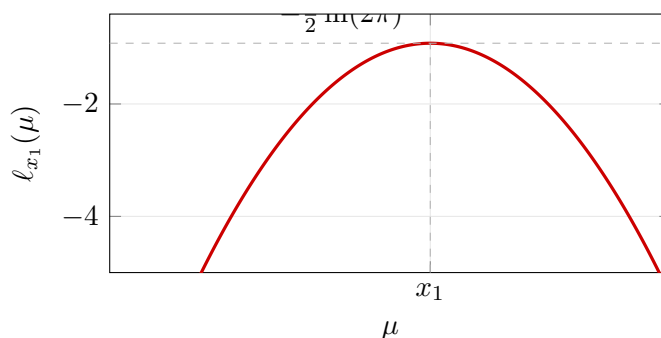
Pre-likelihood $L_{x_1}(\mu) = \varphi(x_1 - \mu)$ vs. μ (red slice), $x_1 = 1$



Taking the log converts the bell into a downward parabola, whose vertex is still at $\mu^* = x_1$:

$$\ell_{x_1}(\mu) = \log \varphi(x_1 - \mu) = -\frac{1}{2}(x_1 - \mu)^2 - \frac{1}{2} \log(2\pi).$$

Log-likelihood $\ell_{x_1}(\mu)$ vs. μ , $n = 1$, $x_1 = 1$



The MLE for $n = 1$ is $\hat{\mu} = x_1$: the single observation is the maximum-likelihood guess for the location.

Case $n = 2$: the 3D picture and its limitation. For two observations (x_1, x_2) the joint log-likelihood is

$$\ell(x_1, x_2; \mu) = -\frac{1}{2}(x_1 - \mu)^2 - \frac{1}{2}(x_2 - \mu)^2 - \ln(2\pi),$$

which was visualised as a downward paraboloid in (x_1, x_2, ℓ) -space in Step 9 (the red bowl on page 41). However, that picture is a function of all three variables simultaneously: it is *not* useful for visualising the MLE, which requires holding (x_1, x_2) fixed and varying μ alone.

Case $n \geq 1$: reduction to $Q(\mu)$. For a general i.i.d. sample $(x_1, \dots, x_n)$ write out the exponent:

$$Q(x_1, \dots, x_n; \mu) := \sum_{i=1}^n (x_i - \mu)^2 = \sum x_i^2 + n\mu^2 - 2\mu \sum x_i = n(\mu^2 - 2\bar{x}\mu + \bar{x}^2) = \tilde{Q}(n, \mu, \bar{x}, \bar{x}^2),$$

where $\bar{x} = \frac{1}{n} \sum x_i$ and $\bar{x}^2 = \frac{1}{n} \sum x_i^2$. The joint density factors as

$$h(\mathbf{x}; \mu) = C \cdot e^{-\frac{1}{2}Q(\mathbf{x}; \mu)},$$

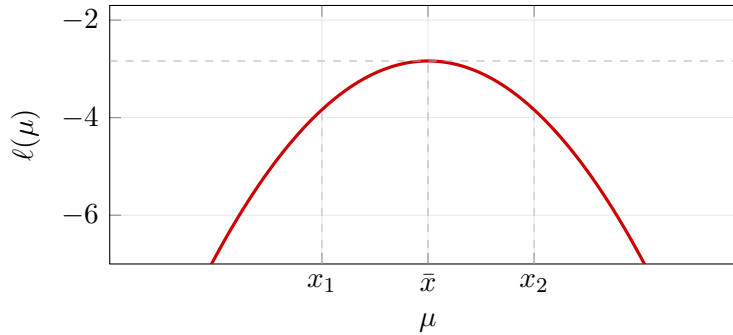
so $\operatorname{argmax}_{\mu} h = \operatorname{argmin}_{\mu} Q$.

Key observation. Although Q depends on the data through *both* $\bar{x}$ and $\bar{x}^2$, the argmin in μ depends only on $\bar{x}$. Concretely, treating Q/n as a function of μ alone:

$$\frac{Q}{n} = \mu^2 - 2\bar{x}\mu + \bar{x}^2, \quad \frac{d}{d\mu} \left(\frac{Q}{n} \right) = 2\mu - 2\bar{x} = 0 \implies \mu^* = \bar{x}.$$

For fixed (x_1, x_2) the log-likelihood $\ell_{x_1, x_2}(\mu)$ is therefore a downward parabola in μ alone, with vertex at $\bar{x} = (x_1 + x_2)/2$:

Log-likelihood vs. μ for fixed $(x_1, x_2) = (0, 2)$; vertex at $\mu^* = \bar{x} = 1$



Summary. The likelihood depends on the data through both $\bar{x}$ and $\bar{x}^2$, but its argmax depends only on $\bar{x}$, and equals $\bar{x}$. This is the sample mean as MLE.

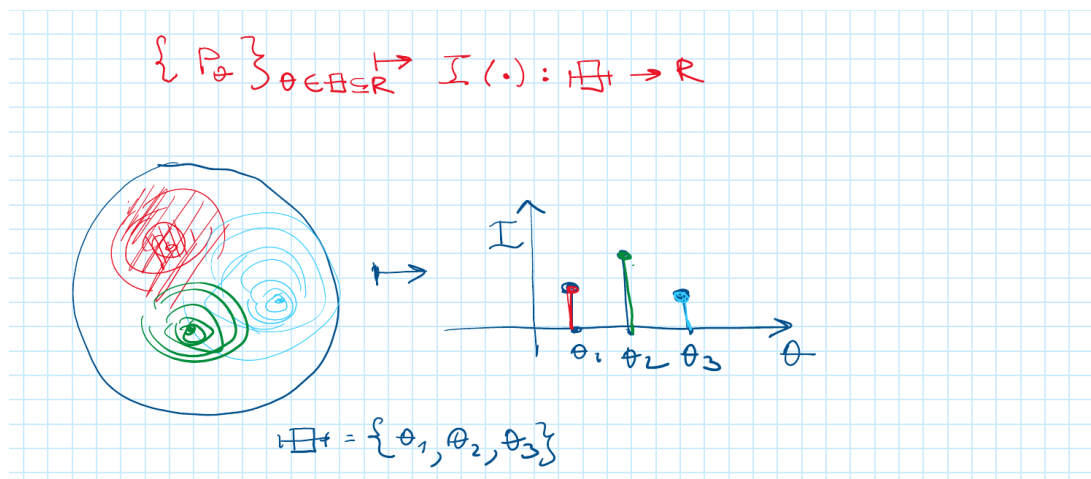
1.4 Fisher information for $\mathcal{N}(\mu, 1)$

Definition. The *Fisher information* is the variance of the score:

$$\mathcal{I}(\theta) = \operatorname{Var}_{\theta}[s(X, \theta)] = \mathbb{E}_{\theta} \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right] = \int_{\mathbb{R}} \left(\frac{\partial}{\partial \theta} \log f(x; \theta) \right)^2 f(x; \theta) dx,$$

where the last equality holds under absolute continuity (Lebesgue density exists). Since $\mathcal{I}(\theta)$ is the expectation of a square, it is always non-negative.

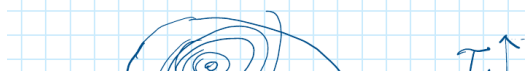
Structurally, $\mathcal{I}$ is an operator on one-parametric families of measures: $\{P_{\theta}\}_{\theta \in \Theta} \mapsto \mathcal{I}(\cdot)$, where $\mathcal{I}(\cdot): \Theta \rightarrow \mathbb{R}$. The parameter space Θ must be a continuum (e.g. an interval) because we need to differentiate in θ . The first picture shows the case of a finite parameter set $\Theta = \{\theta_1, \theta_2, \theta_3\}$ as a toy illustration: a family of measures on the left maps to three scalar values on the right.



When Θ is a continuum the family of measures is harder to depict, but the output function $\mathcal{I}(\cdot)$ is easy — it is just a curve over θ :

Then it will be harder to depict the family of measures on the left.

However, function, defined on continuum, on the right, is easy to depict.



Two routes to $\mathcal{I}$: generic vs. regular. In the generic (absolutely continuous) case:

1. Square the score $s(x, \theta)$.
2. Integrate against the density $f_\theta(x)$.

Under *regularity conditions* (differentiation under the integral sign is valid) a third equivalent formula holds:

$$\mathcal{I}(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \ln L_x(\theta) \right] = -\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} s_x(\theta) \right].$$

This adds a differentiation and a negation step:

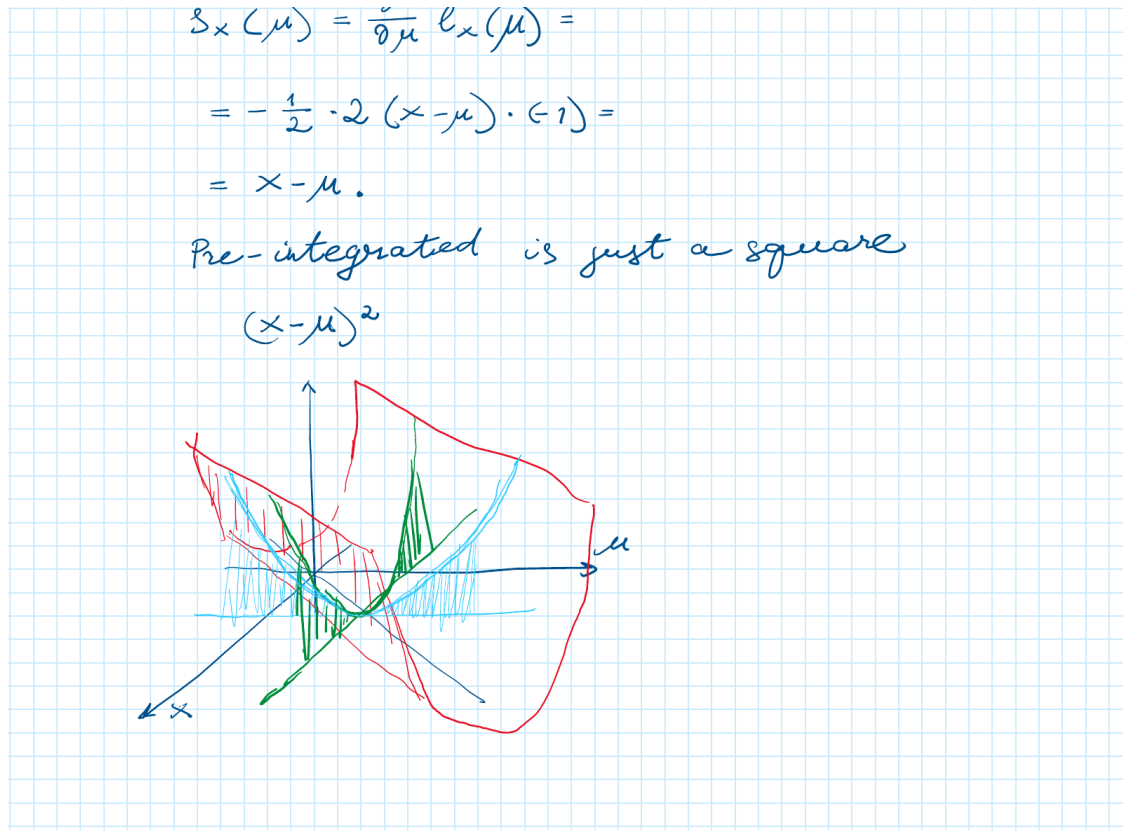
1. Differentiate the score in θ .
2. Integrate against f_θ .
3. Negate.

We work out both for $\mathcal{N}(\mu, 1)$.

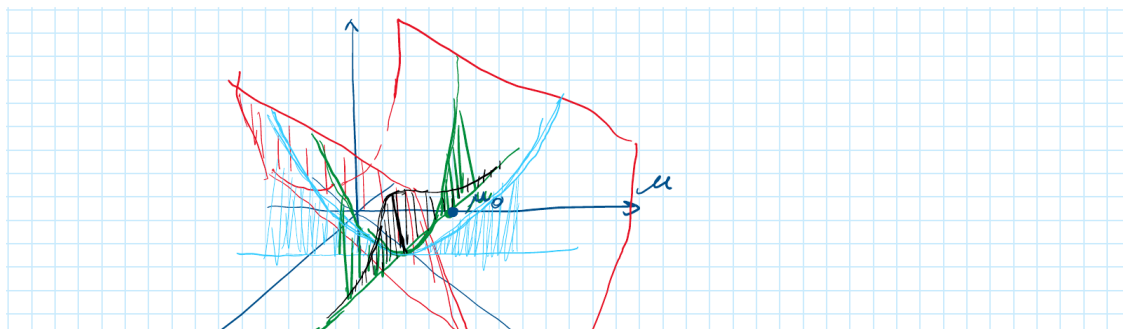
Score for $\mathcal{N}(\mu, 1)$. For the location family $f_\mu(x) = \varphi(x - \mu)$:

$$\ell_x(\mu) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2}(x - \mu)^2, \quad s_x(\mu) = \frac{\partial \ell_x}{\partial \mu} = x - \mu.$$

The pre-integrand for the generic formula is $(x - \mu)^2$ — just a square. The bivariate surface $s(x, \mu) = x - \mu$ over the (x, μ) -plane:

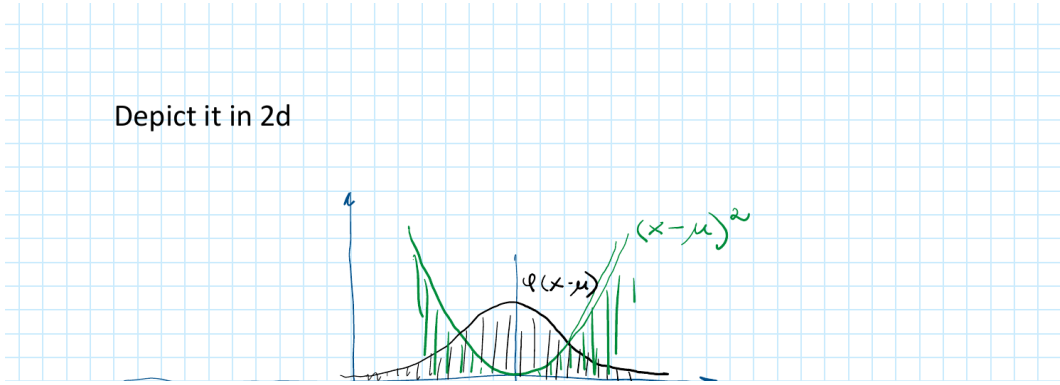


Fix $\mu = \mu_0$. The green density $\varphi(x - \mu_0)$ is added as a section:

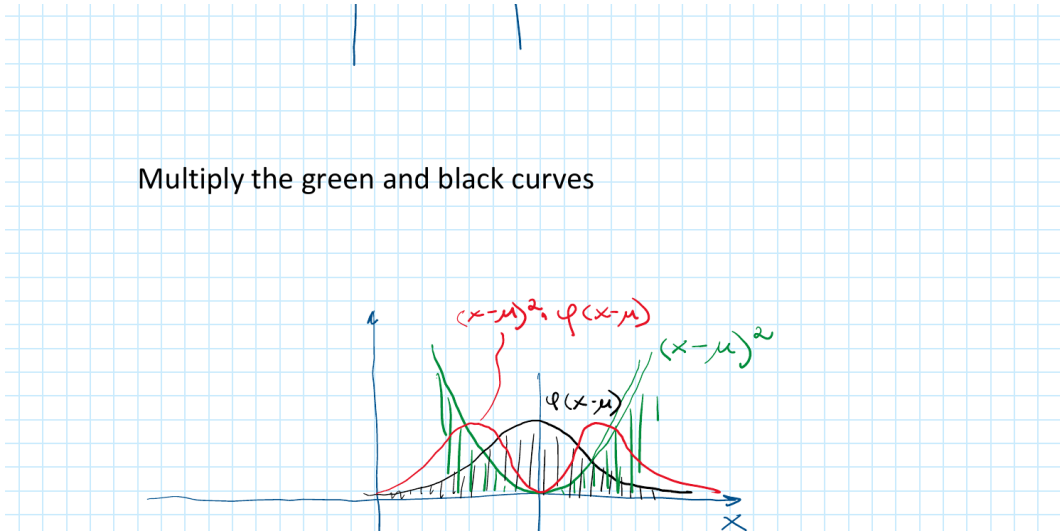


Forming the pre-integral curve. The 2D picture at fixed μ_0 shows the squared score $(x - \mu_0)^2$ (green) and the density $\varphi(x - \mu_0)$ (black) side by side:

Depict it in 2d



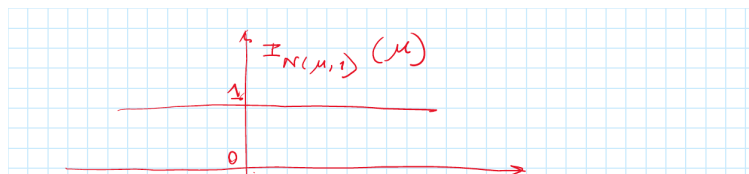
Multiplying them gives the pre-integral curve (red): $(x - \mu_0)^2 \cdot \varphi(x - \mu_0)$:



Computing $\mathcal{I}_{\mathcal{N}(\mu,1)}(\mu)$. Integrating with the substitution $y = x - \mu$:

$$\mathcal{I}_{\mathcal{N}(\mu,1)}(\mu) = \int_{-\infty}^{\infty} (x - \mu)^2 \varphi(x - \mu) dx = \int_{-\infty}^{\infty} y^2 \varphi(y) dy = \text{Var}(Z) = 1, \quad Z \sim \mathcal{N}(0,1).$$

The result does not depend on μ — $\mathcal{I}$ is constant for this family:



This constancy is specific to the location-normal family. In general $\mathcal{I}(\theta)$ depends on θ .

Regular formula: differentiating the score. For the regular formula we need $\frac{\partial}{\partial \mu} s(x, \mu)$. Here $s(x, \mu) = x - \mu$, so:

$$\frac{\partial}{\partial \mu} s(x, \mu) = \frac{\partial}{\partial \mu} (x - \mu) = -1.$$

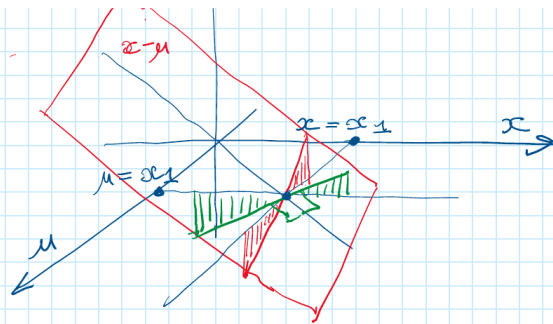
Already at this level the derivative does not depend on x . The 2D section at a fixed μ (the score as a function of x):

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$$s(x, \mu) = x - \mu$$

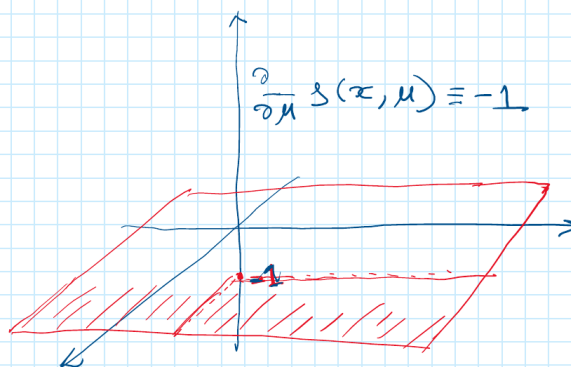


The bivariate surface $\partial_\mu s(x, \mu) \equiv -1$ is a flat horizontal plane at height -1 over the entire (x, μ) -plane:



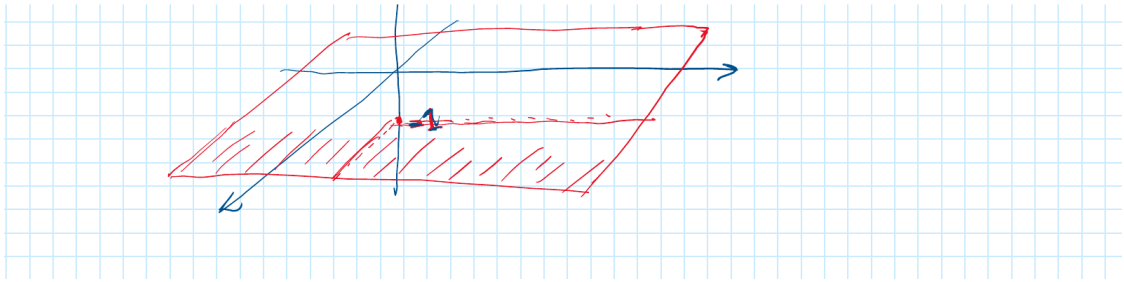
$$\frac{\partial}{\partial \mu} s(x, \mu) = \frac{\partial}{\partial \mu} (x - \mu) = -1$$

Already at this level it does not depend on x.

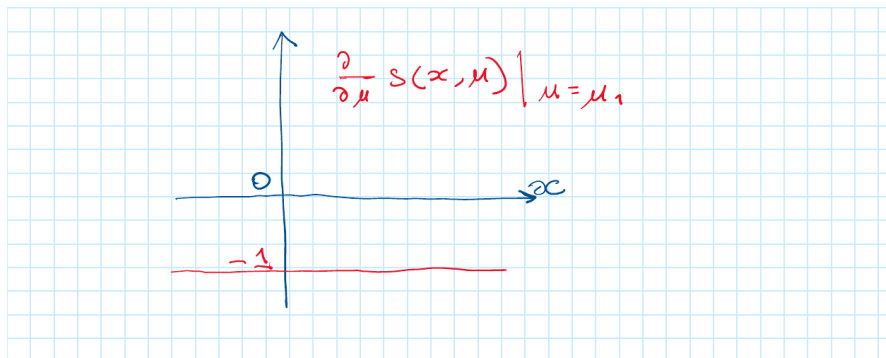


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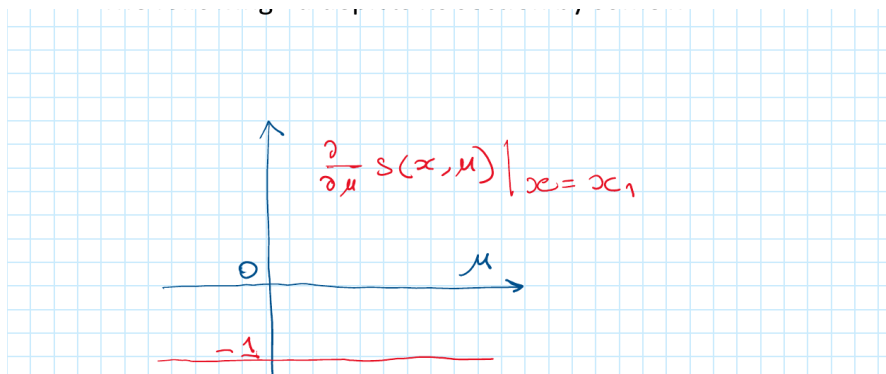
Sections of $\partial_\mu s \equiv -1$. The flat plane is also shown alone, then its 2D sections:



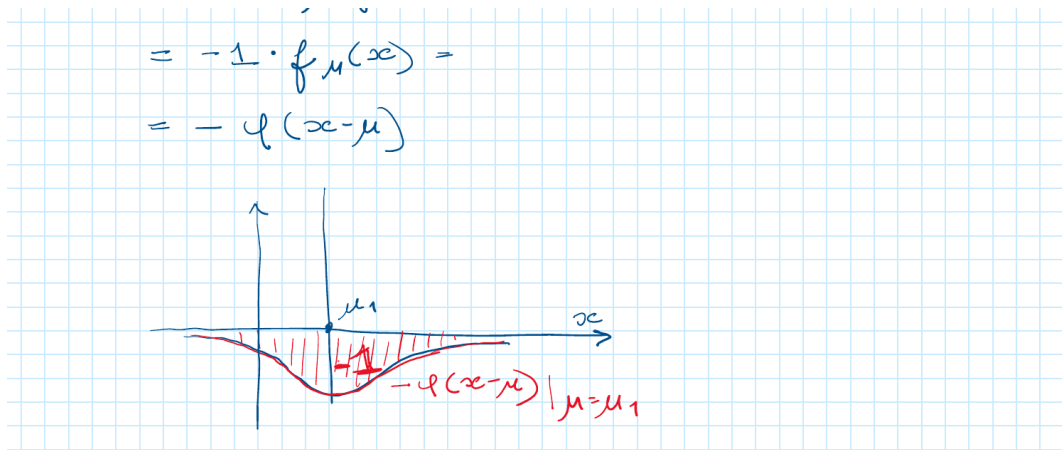
Section at fixed $\mu = \mu_1$ (as a function of x): a horizontal line at -1 :



Section at fixed $x = x_1$ (as a function of μ): also a horizontal line at -1 :

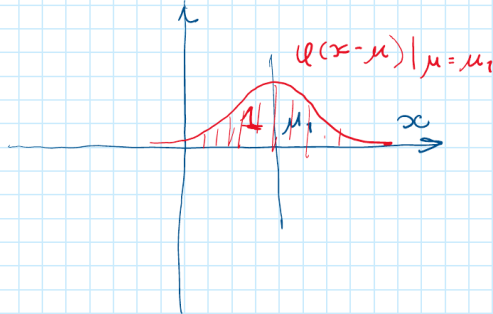


Forming the product and negating. The product $\frac{\partial}{\partial \mu} s(x, \mu) \cdot f_\mu(x) = (-1) \cdot \varphi(x - \mu) = -\varphi(x - \mu)$ at $\mu = \mu_1$ is the negative density — an upside-down bell curve:



Negating gives $\varphi(x - \mu)$, which integrates to 1 by Lebesgue:

And it integrates to 1 by Lebesgue.



$$\mathcal{I}(\mu) = - \int_{-\infty}^{\infty} \frac{\partial}{\partial \mu} s(x, \mu) \cdot \varphi(x - \mu) dx = \int_{-\infty}^{\infty} \varphi(x - \mu) dx = 1.$$

Both routes confirm $\mathcal{I}_{\mathcal{N}(\mu, 1)} \equiv 1$.